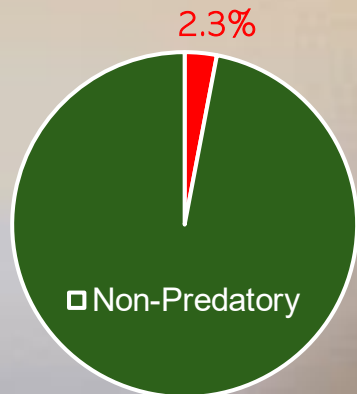
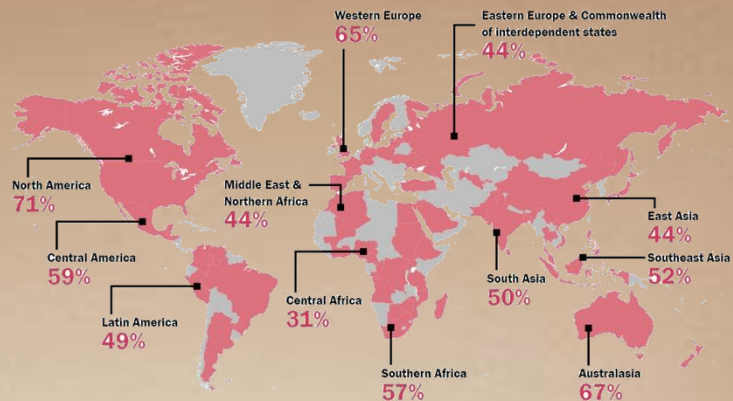




Online Grooming

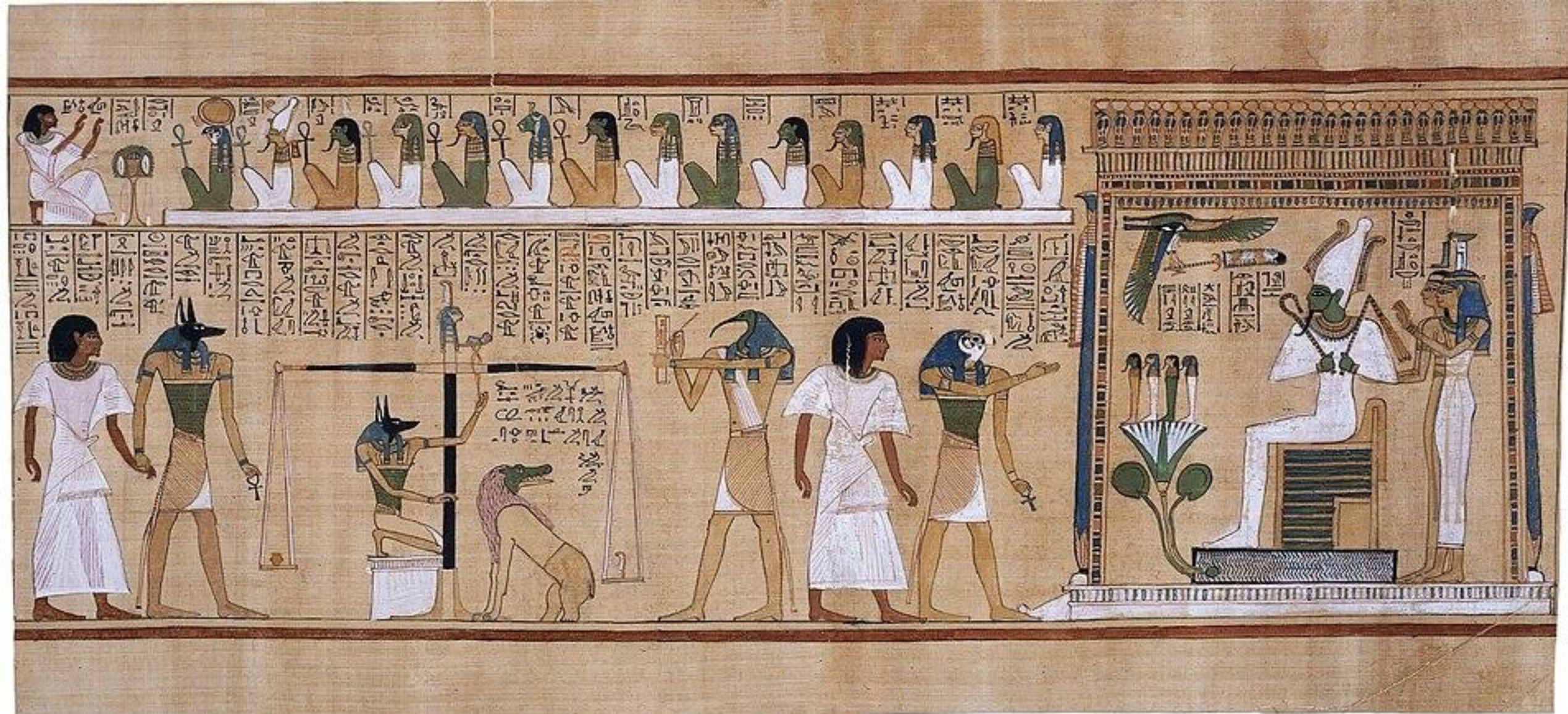


Osprey: A Reference Framework for Online Grooming Detection

<https://github.com/fani-lab/Osprey>

Outline: W05

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- IIII) Evaluation**
Metrics, Methodology



Duat /doo·et/ or Tuat /taat/ where people go after they die, in ancient Egyptian mythology. The '*Weighing of the Heart*' from the '*Book of the Dead*' from the Papyrus of Hunefer, dated to the 19th Dynasty around 1275 BCE. The deceased Hunefer is taken into the judgment hall by the deity Anubis, who weighs a portion of Hunefer's soul, represented by his heart. This ritual is completed with Ammit the Devourer awaiting the result, and Thoth recording. Next, the triumphant Hunefer, having passed the test, is presented by the falcon-headed Horus to Osiris, seated in his shrine with Isis, Nephthys and the four sons of Horus.

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Fully Supervised Classification Evaluation

Technique (k NN, LR, SVM, ...) does *not* matter ...

Students (technique) learn from the course (labeled set) differently, yet all are evaluated/judged the same.



Statue of Lady Justice blindfolded and holding a balance and a sword, outside the [Court of Final Appeal, Hong Kong](#)

$$(o, c) \rightarrow f(o) \cong c$$

Lady Justice, or the goddess Justitia, an allegorical personification of justice and moral force in ancient Roman art known as Iustitia or Justitia. Her attributes are scales, a sword and blindfold.

https://en.wikipedia.org/wiki/Lady_Justice

Outline: W05

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- IIII) Evaluation
 - Accuracy

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Fully Supervised Classification Evaluation

Accuracy

$$(o, c) \rightarrow f(o) \cong c$$

$$(q, a) \rightarrow f(q) \cong a$$

For many (o, c) , how often $f(o) = c$...

For many (q, a) in the exam, how often $Hossein(q) = a$...

For a random (o, c) , what's the probability prior to answer: $\mathbf{P}(f(o) = c \mid (o, c))$

For a random (q, a) , what's the probability (before exam) $Hossein(q) = a$? $\mathbf{P}(Hossein(q)=c \mid (o, c))$

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Fully Supervised Boolean Classification Evaluation

Accuracy

$$(o, 1) \rightarrow f(o) \cong 1$$

$$(o, 0) \rightarrow f(o) \cong 0$$

For many (o, c) , how often $f(o) = 1$ if $c = 1$, and how often $f(o) = 0$ if $c = 0$...

Being in a binary choice (True/False) exam ...

For a random (o, c) , what's the probability

A random question from the pool of binary choice questions

$$\mathbf{P}(f(o) = c \mid (o, c)) = \mathbf{P}(f(o) = 1) \times \mathbf{P}(c = 1) + \mathbf{P}(f(o) = 0) \times \mathbf{P}(c = 0)$$

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Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

	Accuracy
A biased pessimist $f(o) = 1$ unless proved otherwise!	0.5
A biased optimist $f(o) = 0$ unless proved otherwise.	0.5
An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!	0.5
An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...	0.5
A data mining technique (method)	0.50000001 at least
The Oracle of Delphi	1.0

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Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced (unbalanced) distribution of classes

$$(o, 1)_{001 \dots 001} \rightarrow f(o) \cong 1$$

$$(o, 0)_{002 \dots 100} \rightarrow f(o) \cong 0$$

Email spam detection

Credit-card fraud detection

Medical diagnosis

Loan default prediction

Customer churn

Network intrusion detection

Manufacturing defect detection

Insurance claim fraud

Ad click prediction

Fake review detection

Cancer screening

Majority class

Not spam

Legitimate

Healthy

Repay

Stays

Normal traffic

No defect

Legitimate

Does not click

Genuine

Negative

Minority class

Spam

Fraudulent

Disease

Defaults

Churns

Attack

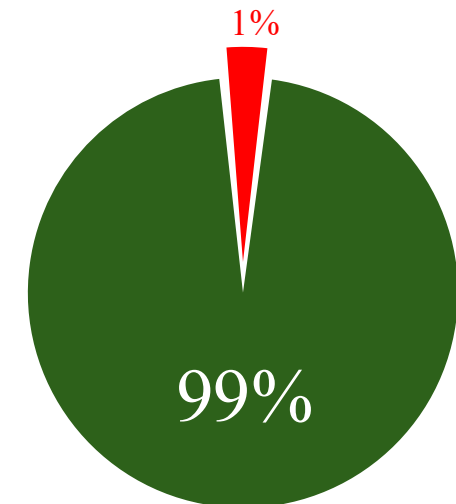
Defective

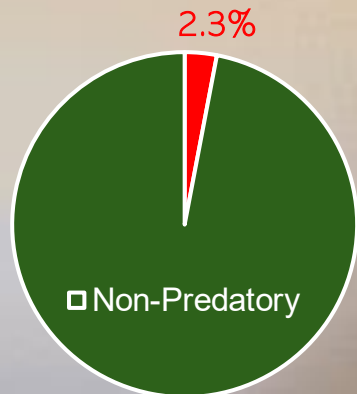
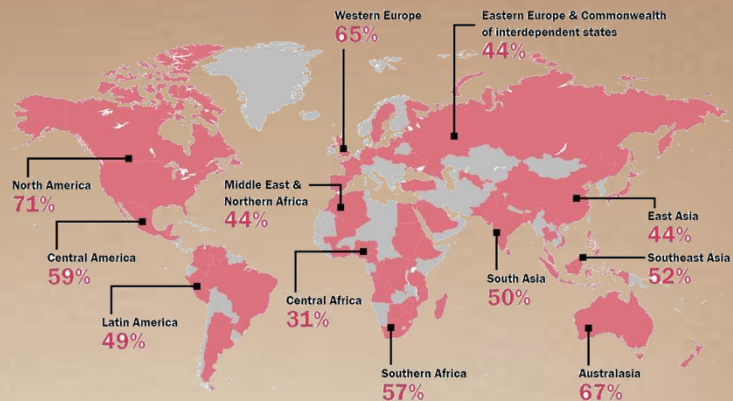
Fraudulent

Clicks

Fake

Positive





Online Grooming



Osprey: A Reference Framework for Online Grooming Detection

<https://github.com/fani-lab/Osprey>

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Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \dots 001} \rightarrow f(o) \cong 1$$

$$(o, 0)_{002 \dots 100} \rightarrow f(o) \cong 0$$

Accuracy

A biased pessimist $f(o) = 1$ unless proved otherwise!

?

A biased optimist $f(o) = 0$ unless proved otherwise.

?

An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!

?

An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...

?

A data mining technique (method)

? at least

The Oracle of Delphi

1.0

Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \dots 001} \rightarrow f(o) \cong 1$$

$$(o, 0)_{002 \dots 100} \rightarrow f(o) \cong 0$$

Accuracy

A biased pessimist $f(o) = 1$ unless proved otherwise!

0.01 (poor but savior)

A biased optimist $f(o) = 0$ unless proved otherwise.

0.99 (accurate but useless)

An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!

?

An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...

?

A data mining technique (method)

? at least

The Oracle of Delphi

1.0

Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \dots 001} \rightarrow f(o) \cong 1$$

$$(o, 0)_{002 \dots 100} \rightarrow f(o) \cong 0$$

Accuracy

A biased pessimist $f(o) = 1$ unless proved otherwise!

0.01 (poor but savior)

A biased optimist $f(o) = 0$ unless proved otherwise.

0.99 (accurate but useless)

An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!

0.50

An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...

0.50

$$P(f(o) = c) = P(f(o) = 1) \times P(c = 1) + P(f(o) = 0) \times P(c = 0)$$

$$= P(f(o) = 1) \times \frac{1}{100} + P(f(o) = 0) \times \frac{99}{100} \text{ w/ equal probabilities to guess 1 or 0, so } \frac{1}{2} \text{ each} = \frac{1}{2} \times \frac{1}{100} + \frac{1}{2} \times \frac{99}{100} = 0.5$$

Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \dots 001} \rightarrow f(o) \cong 1$$

$$(o, 0)_{002 \dots 100} \rightarrow f(o) \cong 0$$

Accuracy

A biased pessimist $f(o) = 1$ unless proved otherwise!

0.01 (poor but savior)

A biased optimist $f(o) = 0$ unless proved otherwise.

0.99 (accurate but useless)

An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!

0.50

An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...

0.50

A data mining technique (method)

0.99000001 at least

The Oracle of Delphi

1.0

Fully Supervised Boolean Classification Evaluation

Accuracy: Imbalanced distribution of classes

$$(o, 1)_{001 \dots 001} \rightarrow f(o) \cong 1$$

$$(o, 0)_{002 \dots 100} \rightarrow f(o) \cong 0$$

Accuracy

A biased pessimist $f(o) = 1$ unless proved otherwise!

0.01 (poor but savior)

A biased optimist $f(o) = 0$ unless proved otherwise.

0.99 (accurate but useless)

Not a good metric for unbalanced classes; gives low score to a useful model and high score to a useless!

→ Because it gives **equal importance to inequal classes!**

→ We need to focus evaluating f for minority class ...

→ Other metrics ...



Outline: W05

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- IIII) Evaluation**
 - Confusion Matrix

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Fully Supervised Boolean Classification Evaluation

https://en.wikipedia.org/wiki/Precision_and_recall

		True condition				
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$	
Predicted condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$	
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$	$F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate = $\frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$		

Fully Supervised Boolean Classification Evaluation

Contingency Table, Confusion Matrix

		True condition				
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$	
Predicted condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$	
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$	$F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate = $\frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$		

Fully Supervised Boolean Classification Evaluation

The Oracle of Delphi in a balanced distribution of classes

		True condition				
Total population		Condition positive	Condition negative		Condition positive Total population	Accuracy (ACC) = $\frac{\sum \text{ True positive} + \sum \text{ True negative}}{\sum \text{ Total population}}$
Predicted condition	Predicted condition positive	True positive 50	False positive, 0		Positive predictive value (PPV), Precision = $\frac{\sum \text{ True positive}}{\sum \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{ False positive}}{\sum \text{ Predicted condition positive}}$
	Predicted condition negative	False negative, 0	True negative 50		False negative rate (FOR) = $\frac{\sum \text{ False negative}}{\sum \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{ True negative}}{\sum \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{ True positive}}{\sum \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{ False positive}}{\sum \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$	$F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{ False negative}}{\sum \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{ True negative}}{\sum \text{ Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$		

Fully Supervised Boolean Classification Evaluation

A pessimist, an optimist, a drunkard, a gambler

		True condition				
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$	
Predicted condition	Predicted condition positive	True positive ?	False positive ?	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$	
	Predicted condition negative	False negative ?	True negative ?	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$	$F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$		

Fully Supervised Boolean Classification Evaluation

A pessimist in a balanced distribution of classes

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive 50	False positive 50	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative 0	True negative 0	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Fully Supervised Boolean Classification Evaluation

An **optimist** in a balanced distribution of classes

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
Predicted condition	Predicted condition positive	True positive 0	False positive, 0	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$
	Predicted condition negative	False negative, 50	True negative 50	False omission rate (FOR) = $\frac{\Sigma \text{False negative}}{\Sigma \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\Sigma \text{True positive}}{\Sigma \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{False positive}}{\Sigma \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{False negative}}{\Sigma \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{True negative}}{\Sigma \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Fully Supervised Boolean Classification Evaluation

A drunkard/gambler in a balanced distribution of classes

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive 25	False positive 25	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative 25	True negative 25	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Fully Supervised Boolean Classification Evaluation

The Oracle of Delphi in an **im**balanced distribution of classes (1 vs. 99)

		True condition				
Total population		Condition positive	Condition negative		Condition positive Total population	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive 1	False positive, 0		Positive predictive value (PPV), $\text{Precision} = \frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative, 0	True negative 99		False negative rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$	$F_1 \text{ score} =$ $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$		

Fully Supervised Boolean Classification Evaluation

A pessimist, optimist, drunkard, gambler in an **im**balanced distribution of classes (1 vs. 99)

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive ?	False positive, ?	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative, ?	True negative ?	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Fully Supervised Boolean Classification Evaluation

A pessimist in an **im**balanced distribution of classes (1 vs. 99)

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
Predicted condition	Predicted condition positive	True positive 1	False positive, 99	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$
	Predicted condition negative	False negative, 0	True negative 0	False omission rate (FOR) = $\frac{\Sigma \text{False negative}}{\Sigma \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\Sigma \text{True positive}}{\Sigma \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{False positive}}{\Sigma \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{False negative}}{\Sigma \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{True negative}}{\Sigma \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Fully Supervised Boolean Classification Evaluation

An **optimist** in an **im**balanced distribution of classes (1 vs. 99)

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive 0	False positive, 0	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative, 1	True negative 99	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Fully Supervised Boolean Classification Evaluation

A drunkard/gambler in an imbalanced distribution of classes (1 vs. 99)

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive ?	False positive, ?	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative, ?	True negative ?	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

A challenging sample exam question ...

Outline: W05

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

IIII) Evaluation

Confusion Matrix → A Single Score: Precision

Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$

From the model's perspective ...

Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$

From the model's perspective in a balanced dataset (50 vs. 50) ...

- Pessimist: $\frac{TP=50}{100} = 0.5$
- Optimist: Never says positive $\frac{TP=0}{0} \rightarrow 0.0$
- Drunk/Gambler: $\frac{TP=25}{50} = 0.5$

Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$

From the model's perspective in an imbalanced dataset (1 vs. 99) ...

- Pessimist: $\frac{TP=1}{100} = 0.1$ (but a valuable model, catches that rare positive)
- Optimist: Never says positive $\frac{TP=0}{0} \rightarrow 0.0$
- Drunk/Gambler: $\frac{TP=?}{?} = ?$

Fully Supervised Boolean Classification Evaluation

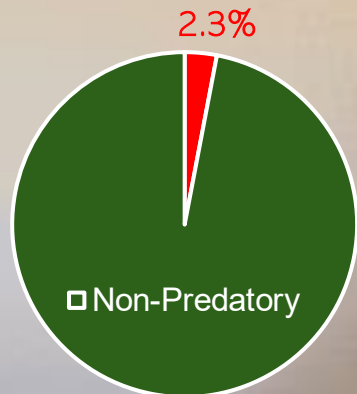
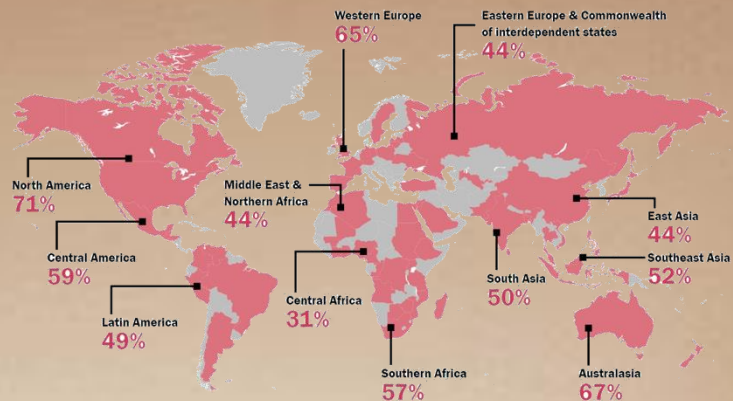
Precision: How many of the model's catches (positive) are correct?

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$

From the model's perspective ...

What scenarios/applications require **high** precision?

Better question would be "what happens if we have **low** precision? Will it be **disaster**?"



Online Grooming



Osprey: A Reference Framework for Online Grooming Detection

<https://github.com/fani-lab/Osprey>

Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$

The observations are online users.

The classification task is catching a child abuser (predator).

The classifier is a Boolean alarm or notifier.

It infers (predicts) positive class! That is an abuser, **notify child or parent!**

From the alarm perspective, what happens if it notify with low precision?

Majority of the notifications are false. Not a disaster. We'll check users one by one.

Fully Supervised Boolean Classification Evaluation

Precision: How many of the model's catches (positive) are correct?

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$

The observations are flying objects.

The classification task is catching/attacking enemy fighter jets.

The classifier is a Boolean missile system.

It infers (predicts) positive class! That is an enemy fighter jet approaching, **fire!**

From the missile system perspective, what happens if it fires with low precision?



Ukraine International Airlines Flight 752

https://en.wikipedia.org/wiki/Ukraine_International_Airlines_Flight_752

